

# Md Marufi Rahman, Ph.D.

(940) 268-4062 | [mm\\_rahman\\_111@outlook.com](mailto:mm_rahman_111@outlook.com) | [Google Scholar](#) | Dallas, TX  
[LinkedIn: mdmarufirahman](#) | [GitHub: mdmarufirahman.github.io](#)

Work Authorization: U.S. Permanent Resident — Authorized to work for any employer

## RESEARCH SUMMARY

---

Ph.D. in Computer Science with 6+ years of experience building AI/ML systems for safety-critical applications in medical imaging, multimodal retrieval, and high-performance computing. Research focuses on reliable, interpretable deep learning — training and deploying models that need to work correctly under real-world distribution shifts, not just on clean benchmarks.

## TECHNICAL SKILLS

---

**Languages:** Python, C++, CUDA, MATLAB, SQL, Bash

**AI/ML:** PyTorch, TensorFlow, Hugging Face Transformers, ONNX, OpenCV, MediaPipe, LangChain, FastAPI

**HPC & Infrastructure:** CUDA/GPU optimization, Nsight profiling, AWS, GCP, Azure, Docker, Kubernetes, CI/CD, Linux

**Retrieval & GenAI:** RAG pipelines, FAISS, Pinecone, OpenSearch, BLIP-2, CLIP, Diffusion Models, Agentic AI Workflows

## RESEARCH EXPERIENCE

---

### Graduate Research Assistant — Machine Learning & Computer Vision

Aug 2020 – May 2026

Multimedia Information Group, University of North Texas | Denton, TX

- Developed a content-based retrieval system for colonoscopy imagery combining monocular depth estimation with geometric priors, improving structural matching F1-score to 92.5% and reducing redundant retrievals; published as first author at BIOSTEC 2023.
- Built multi-GPU deep learning pipelines for medical video analysis in PyTorch and CUDA, achieving 4x training speedup and 2.5x lower inference latency over prior TensorFlow baselines.
- Implemented GPU-accelerated motion estimation and sensor-processing routines in CUDA, tuned with Nsight profiling, reaching 2.2x throughput over CPU baselines for real-time deformation tracking.
- Fine-tuned BLIP-2 vision-language models for medical visual question answering, improving ROUGE-L by 12% and reducing annotation requirements through uncertainty-based active sampling.
- Deployed models as containerized inference services using FastAPI, Docker, and ONNX with automated monitoring and model-drift alerting, sustaining sub-100 ms p95 latency.
- Built retrieval-augmented search assistants with LangChain, OpenSearch, and vector databases (FAISS, Pinecone) for document and experiment-log analysis across research corpora.
- Mentored 3 junior researchers on deep learning practice, GPU optimization, and code review, contributing to 2 peer-reviewed publications.

## TEACHING EXPERIENCE

---

### Teaching Fellow

Aug 2019 – Aug 2025

Department of Computer Science & Engineering, University of North Texas | Denton, TX

- Instructed 40+ students per semester across graduate and undergraduate courses in databases, algorithms, and data structures.
- Designed hands-on assignments in Python, C++, and SQL; provided 1-on-1 mentorship and collaborated with faculty on course materials and grading.

## EDUCATION

---

### Ph.D., Computer Science — University of North Texas, Denton, TX

Jan 2023 – May 2026

GPA: 3.85 / 4.0. Dissertation: robust and interpretable deep learning for medical imaging and multimodal retrieval.

### M.S., Computer Science — University of North Texas, Denton, TX

Aug 2019 – Dec 2022

## SELECTED PUBLICATIONS

---

- **M. M. Rahman**, J. Oh, and P. C. de Groen, “Non-Informative Frame Classification in Colonoscopy,” in Proc. 30th Int. Conf. Knowledge-Based and Intelligent Information and Engineering Systems (KES 2026), Sep. 2026 (under review).
- J. I. M. Rifat, T. Tabashum, **M. M. Rahman**, M. F. Mokter, S. Engala, and S. Bozdag, “BioLM-NET: an interpretable deep learning model combining prior biological knowledge and contextual LLM gene embeddings on multi-omics data to predict disease,” in Biocomputing 2026, Kohala Coast, Hawaii, USA: WORLD SCIENTIFIC, Dec. 2025, pp. 644–663. doi: 10.1142/9789819824755\_0047.
- **M. M. Rahman**, J. Oh, W. Tavanapong, and P. C. de Groen, “Content Based Image Retrieval Using Depth Maps for Colonoscopy Images,” in Proc. 16th Int. Joint Conf. on Biomedical Engineering Systems and Technologies (BIOSTEC 2023), Lisbon, Portugal: SCITEPRESS, 2023, pp. 301–308. doi: 10.5220/0011749100003414.
- Y. Li, J. Baik, **M. M. Rahman**, I. Anagnostopoulos, R. Li, and T. Shu, “Pareto Optimization of CNN Models via Hardware-Aware Neural Architecture Search for Drainage Crossing Classification on Resource-Limited Devices,” in Proc. SC '23 Workshops of the Int. Conf. on High Performance Computing, Network, Storage, and Analysis, Denver, CO, USA: ACM, Nov. 2023, pp. 1767–1775. doi: 10.1145/3624062.3624258.
- **M. M. Rahman**, J. Oh, W. Tavanapong, J. Wong, and P. C. De Groen, “Automated Bite-block Detection to Distinguish Colonoscopy from Upper Endoscopy Using Deep Learning,” in Advances in Visual Computing (Lecture Notes in Computer Science, vol. 13018), Cham: Springer, 2021.

## SELECTED AWARD

---

- NSF Scholarship Recipient — Deep Learning Systems & Advanced GPU Cyberinfrastructure Workshop (2023). Competitively awarded to select graduate researchers nationwide.